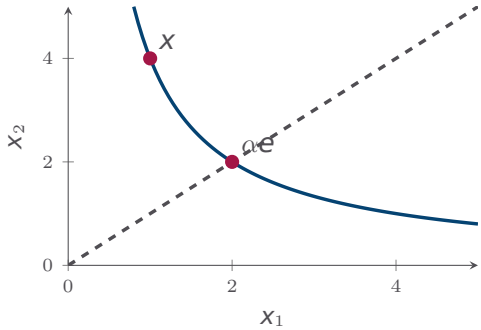
A constructive picture: the diagonal bundle



Now add monotonicity to rationality and continuity. Each x is indifferent to a unique diagonal bundle αe, where $e = (1, \ldots, 1)$.

$$u(x) = \alpha \quad \text{if } x \sim \alpha e.$$

A useful way to **construct** a utility scale.

→ Proof under monotonicity